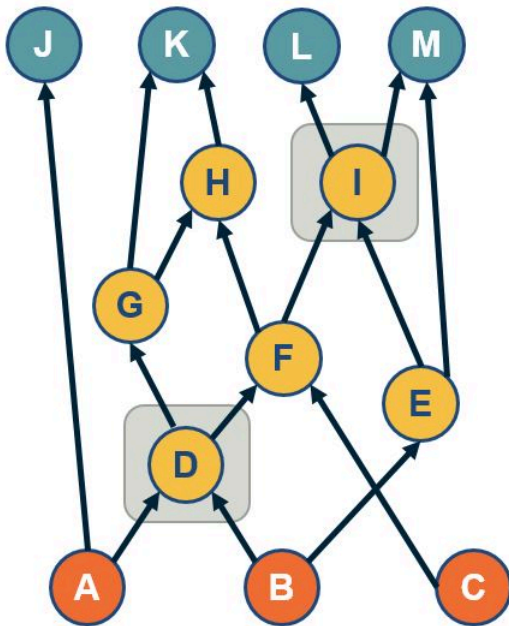


L6: Core Set of Nodes in DAGs



A path-based approach to identify a group of core nodes in a network is referred to as the “ τ -core”.

We define that a node v “covers” a path p when v is traversed by p .

The τ -core is defined as follows: *given a set of network paths P we are interested in, what is the minimal set of nodes that can cover at least a fraction τ of all paths in P ?*

In the context of DAGs, the set P can be the ST-paths from the set of all sources (*orange nodes*) to all targets (*blue nodes*).

The rationale for covering only a fraction τ of paths (say 90%) instead of all paths is because in many real networks there are some ST-paths that do not traverse any intermediate nodes.

The problem of computing the the “ τ -core” has been shown to be NP-Complete in the following paper. It can be approximated by a greedy heuristic that iteratively selects the node with the maximum number of remaining uncovered paths until the τ constraint is met.

Sabrin, Kaeser M., and Constantine Dovrolis. “The hourglass effect in hierarchical dependency networks.” *Network Science* 5.4 (2017): 490-528.

The visualization shows the two core nodes (d and i) for the value of $\tau=80\%$. Note that there are three ST-paths ($a-j$, $b-e-m$ and $c-f-h-k$) that are not covered by this core set.